Laboratorio de Microeconomía I

Centro de Investigación y Docencia Económicas

Maestría en Economía - 2025

Laboratorista: Arturo López

Tarea 13

Exercises

Ex. 4.6 A firm j in a competitive industry has total cost function $c^j(q) = aq + b_j q^2$, where $a > 0$, q is firm output, and b_j is different for each firm.

(a) If $b_j > 0$ for all firms, what governs the amount produced by each of them? Will they produce equal amounts of output? Explain.

(b) What happens if $b_j < 0$ for all firms?

Ex. 4.7 Technology for producing q gives rise to the cost function $c(q) = aq + bq^2$. The market demand for q is $p = \alpha - \beta q$.

(a) If $a > 0$, if $b < 0$, and if there are J firms in the industry, what is the short-run equilibrium market price and the output of a representative firm?

(b) If $a > 0$ and $b < 0$, what is the long-run equilibrium market price and number of firms? Explain.

(c) If $a > 0$ and $b > 0$, what is the long-run equilibrium market price and number of firms? Explain.

Ex. 4.22 A monopolist faces linear demand $p = \alpha - \beta q$ and has cost $C = cq + F$, where all parameters are positive, $\alpha > c$, and $(\alpha - c)^2 > 4\beta F$.

(a) Solve for the monopolist's output, price, and profits.

(b) Calculate the deadweight loss and show that it is positive.

(c) If the government requires this firm to set the price that maximises the sum of consumer and producer surplus, and to serve all buyers at that price, what is the price the firm must charge? Show that the firm's profits are negative under this regulation, so that this form of regulation is not sustainable in the long run.

Ex. 4.23 (Ramsey Rule) Building from the preceding exercise, suppose a monopolist faces negatively sloped demand, $p = p(q)$, and has costs $C = cq + F$. Now suppose that the government requires this firm to set a price (p^*) that will maximise the sum of consumer and producer surplus, subject to the constraint that firm profit be non-negative, so that the regulation is sustainable in the long run. Show that under this form of regulation, the firm will charge a price greater than marginal cost, and that the percentage deviation of price from marginal cost ($(p^* - c)/p^*$) will be proportional to $1/\epsilon^*$, where ϵ^* is the elasticity of firm demand at the chosen price and output. Interpret your result.

Ex. 4.26 A competitive industry is in long-run equilibrium. Market demand is linear, $p = a - bQ$, where $a > 0$, $b > 0$, and Q is market output. Each firm in the industry has the same technology with cost function, $c(q) = k^2 + q^2$.

(a) What is the long-run equilibrium price? (Assume what is necessary of the parameters to ensure that this is positive and less than a.)

(b) Suppose that the government imposes a per-unit tax, $t > 0$, on every producing firm in the industry. Describe what would happen in the long run to the number of firms in the industry. What is the post-tax market equilibrium price? (Again, assume whatever is necessary to ensure that this is positive and less than a.)

(c) Calculate the long-run effect of this tax on consumer surplus. Show that the loss in consumer surplus from this tax exceeds the amount of tax revenue collected by the government in the post-tax market equilibrium.

(d) Would a lump-sum tax, levied on producers and designed to raise the same amount of tax revenue, be preferred by consumers? Justify your answer.

(e) State the conditions under which a lump-sum tax, levied on consumers and designed to raise the same amount of revenue, would be preferred by consumers to either preceding form of tax.

Ex. 4.27 A per-unit tax, $t > 0$, is levied on the output of a monopoly. The monopolist faces demand, $q = p^{-\epsilon}$, where $\epsilon > 1$, and has constant average costs. Show that the monopolist will increase price by more than the amount of the per-unit tax.